

ANALYZING CUSTOMER SPENDING USING MULTIPLE LINEAR REGRESSION

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ABSTRACT

This study analyses customer spending using Multiple Linear Regression modeling. The research utilized a customer analytics dataset obtained from Kaggle, containing demographic and financial variables including age, gender, annual income, credit score, estimated savings, preferred spending categories and loyalty years. The primary objective was to develop and validate a Multiple Linear Regression model that quantifies the relationship between these customer characteristics and spending scores. The methodology involved data pre-processing, exploratory data analysis, model development and comprehensive diagnostic testing of regression assumptions including linearity, independence, homoscedasticity and multicollinearity. Model's goodness of fit were evaluated using R-squared and adjusted R-squared while model performance were evaluated using mean absolute error and root mean square error metrics. The results showed that the model explains approximately 91.7% of the variation in customer spending scores ($R^2 = 0.917$; Adjusted $R^2 = 0.913$), indicating strong explanatory power. Predictive performance on the test dataset showed a Mean Squared Error (MSE) of 78.185, a Root Mean Squared Error (RMSE) of 8.842, and a Mean Absolute Error (MAE) of 7.109, demonstrating low prediction error relative to the 1–100 spending score scale. The results indicate that loyalty years, age group and preferred spending category significantly influence customer spending behavior, while gender, annual income and credit score were not statistically significant predictors. Overall, the findings suggest that behavioral and demographic factors play a more substantial role in determining spending patterns than purely financial characteristics.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the study

The global retail and consumer markets represent a massive economic force with consumer spending accounting for higher spending year by year. According to macrotrends (2024), world consumer spending reached \$60 trillion in 2023, reflecting a steady upward trend from \$56.3 trillion in 2022 and \$53.7 trillion in 2021. Despite this massive financial footprint, businesses and retailers face significant challenges in understanding and predicting customer spending. The ‘customer analytics gap’, the disconnect between available consumer data and actionable business insights remains a critical problem in modern commerce. The complexity of consumer spending patterns stems from multiple interacting factors including demographic characteristics, financial capacity and psychological preferences. Traditional approaches to customer analysis, which often rely on simple demographic segmentation or basic transactional summaries, fail to capture the sophisticated relationships that drive purchasing decisions. This analytical shortfall results in substantial financial losses for businesses through misallocated marketing budgets, ineffective customer retention strategies, and suboptimal product placement.

In the field of information technology, machine learning is one of the fastest-growing fields aiding data scientists in the discovery of hidden patterns in massive datasets. As a foundational supervised machine learning technique, regression analysis provides a powerful framework for this kind of tasks. It examines the effects of variables evaluated on several scales, allowing researchers to determine the optimal set of variables to utilize when developing prediction models. Researchers have employed machine learning techniques to analyze customer spending. Siddique and Wahid (2025) applied the Random Forest algorithm to transactional data to analyze and predict customer spending. Their methodology focused on leveraging a robust, tree-based ensemble method to capture complex patterns in customer. Their findings revealed that the Random Forest model offered high predictive power, effectively handling non-linear relationships within the data to deliver accurate spending forecasts.

For analysis of changes in consumer spending, Regression models have been used. Abdulhussein et al. (2022) demonstrated this by employing an ordered logistic regression model to analyze how demographics influenced whether consumers increased, decreased, or maintained their online spending across 12 product categories during the COVID-19 pandemic. Their results quantitatively confirmed that variables like age, gender, and residence were significant predictors of spending shifts. This study provides robust, empirical evidence for the consistent influence of demographic profiles on spending, even during exceptional economic disruptions. Analyzing customer spending extends beyond using regression models to combination of regression models alongside more complex algorithms. In a comprehensive study, Dang et al. (2023) investigated the feasibility of various methods for predicting customer spending based on past purchase behavior, recognizing this as crucial for business planning and budgeting. Their research specifically evaluated Linear Regression alongside more complex algorithms like Random Forest and LightGBM, establishing Linear Regression as a viable baseline model for spending prediction tasks. While their work explored advanced probabilistic models like BG/NBD and Gamma-Gamma for customer lifetime value, the inclusion and evaluation of Linear Regression validates its relevance for spending prediction problems. Beyond traditional regression and machine learning techniques, innovative database approaches have also been applied to understand consumer spending behavior. Hristoski et al. (2019) proposed a graph database methodology for analyzing financial transactions, focusing on the relationships among transaction counterparts rather than just individual attributes. Their research, implemented in Neo4j using Cypher query language, demonstrated how graph databases could reveal patterns in how people perceive their finances through spending patterns.

Many analysis techniques have been used on customer analytics datasets to analyze the customer spending and others to segment customers based on their spending patterns. But the data that is used varies.

1.2 Statement of the Problem

Accurately analyzing customer spending has become a critical imperative for businesses and financial institutions seeking to enhance marketing strategies, optimize resource allocation and improve customer retention in a competitive global marketplace. The inherent complexity of consumer data influenced by a multitude of interrelated factors such as demographics, income, financial standing and personal preferences poses significant challenges to understanding expenditure patterns. While businesses collect extensive customer data, many lack the statistical framework to quantify how these different factors collectively influence spending behavior. This study addressed this gap by applying Multiple Linear Regression to analyze the relationship between customer attributes and spending patterns. The application of this statistical technique to customer analytics data enables businesses to identify key spending drivers and develop data-driven strategies for revenue growth. This can lead to more targeted marketing campaigns, improved strategic decision-making and commercial outcomes that support both business profitability and enhanced customer satisfaction.

1.3 Objectives of the Study

1.3.1 General objective

To analyse customer spending using Multiple Linear Regression

1.3.2 Specific objectives

- i. To fit a Multiple Linear Regression model to a customer analytics dataset.
- ii. To perform diagnostic checking of the regression model's assumptions
- iii. To evaluate the performance of the model
- iv. To predict customer spending

1.4 Significance of the study

Since the complexity and variability of consumer markets continue to evolve, there is a growing need to study customer spending using statistical modeling techniques. As retail and service sectors expand, understanding and predicting spending patterns becomes increasingly important for businesses, marketers, and financial analysts. The competitive nature of modern commerce requires more accurate analysis of customer spending to help with marketing strategy, customer segmentation, and resource allocation. The availability of comprehensive customer data and advanced statistical methods made possible through Multiple Linear Regression can uncover meaningful patterns and relationships within consumer datasets to provide actionable insights that organizations can use for better decision making. Furthermore, as personalized marketing becomes more prevalent in business strategy, the ability to understand individual spending drivers can help lead to customized service offerings that better meet consumer preferences.

1.5 Scope of the study

This study aims at analyzing customer spending for individual consumers using statistical modeling, specifically the Multiple Linear Regression model. The increasing complexity of consumer spending patterns necessitates robust analytical models that can relate various demographic and financial factors causing substantial variations in customer expenditure. The aim of this study is to evaluate how well Multiple Linear Regression can be constructively applied in identifying and quantifying the key drivers of customer spending.

1.6 Assumptions

1. The secondary data accurately represents the target population and contains valid reliable information.
2. The relationship between the predictor variables and the target variable is linear.
3. The observations in the dataset are independent of one another.
4. The residuals of the regression model are normally distributed and exhibit constant variance (homoscedasticity).
5. There is no perfect multicollinearity among the independent variables.

1.7 Limitations and delimitations

Limitations:

1. The study relies on a pre-existing dataset from Kaggle therefore, the researcher has no control over the original data collection process, measurement methods, or data entry quality.
2. The findings may not be fully generalizable to all consumer populations or market sectors, as spending patterns can vary significantly across different geographical regions and cultural contexts.
3. The Multiple Linear Regression model assumes linear relationships between variables, which may not fully capture any potential non-linear dynamics that exist in the underlying consumer spending patterns.

Delimitations:

1. The study is deliberately confined to the application of Multiple Linear Regression, and other modeling techniques like machine learning algorithms are excluded from the scope.
2. The analysis is limited to the demographic and financial variables within the dataset. Behavioural, psychological, and external economic factors are beyond its scope.
3. The research focuses on model development, statistical interpretation, and validation rather than the implementation of the model in a live business environment.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Customer spending has been key factor in Customer Relationship Management. This chapter reviews related studies and models that explain Customer Relationship Management. The review focuses on existing statistical models and their applicability in understanding the various aspects of Customer Relationship Management, also highlights the gaps in other related studies.

2.2 Related models

Consumer spending has been a crucial part in customer relationship management (CRM). Numerous studies have been conducted on modeling and predicting consumer spending across various economic and business contexts. With the global rise in digital transactions and accessible consumer data, statistical and machine learning models have been increasingly employed to analyze spending patterns and forecast purchasing trends. The literature encompasses a range of regression and machine learning techniques tailored to Customer Relationship Management.

Abdulhussein et al. (2022) used an ordered logistic regression model to analyze how demographics influenced whether consumers increased, decreased, or maintained their online spending across 12 product categories during the COVID-19 pandemic. Their results quantitatively confirmed that variables like age, gender and residence were significant predictors of spending shifts. Beyond predictive modeling for specific customer actions, the strategic importance of Customer Relationship Management has been extensively researched using complex regression model. Soltani et al. (2018), in a study on the organizational impact of CRM, investigated how broader factors like technology, customer orientation, and knowledge management contribute to CRM success. Using Partial Least Squares Structural Equation Modeling (PLS-SEM), they established that the effective use of information technology is a critical driver of successful CRM implementation, which in turn enhances overall organizational performance.

Koley (2025) analyzed Customer Relationship Management (CRM) practices at the State Bank of India, employing a suite of statistical tools including ANOVA, correlation analysis, multiple regression analysis and one-sample t-tests to evaluate the impact of CRM strategies on customer satisfaction and loyalty. The study found significant demographic differences in customer awareness of CRM policies and revealed strong positive correlations between CRM awareness, perception and satisfaction. Critically, the multiple regression analysis identified that customer service is a significant determinant of CRM perception and that e-CRM strategies substantially drive customer satisfaction. In analyzing the digital marketplace, Nisar & Prabhakar (2017) employed regression analysis to investigate the determinants of e-satisfaction and its subsequent impact on consumer expenditure in e-commerce. Their research quantitatively established a significant positive relationship, where a one-unit increase in the American Consumer Satisfaction Index (ACSI) led to a 37.8% increase in consumer spending based on the Consumer Price Index (CPI).

The pursuit of accurate customer spending prediction continues to drive methodological comparisons in recent literature. In a direct evaluation of various algorithms, Prathiba et al. (2024) leveraged a suite of models including Linear Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN) and Lasso Regression to predict customer spending scores on a dataset containing demographic and financial variables, reporting exceptionally high performance metrics for the ensemble methods. Their findings suggested that ensemble methods, particularly Random Forest, excel at predicting customer Spending Scores. The challenge of accurately predicting customer spending scores has also been addressed through the comparison of specialized algorithms. In a focused study, Rajesh & Saravanan (2022) specifically benchmarked a Linear Spline Regression (LSR) algorithm against a Gaussian Mixture Model (GMM) to minimize the Root Mean Square Error (RMSE) in spending score prediction. Their research is particularly instructive because it directly compared a regression-based technique (LSR) with a probabilistic model (GMM), finding that GMM achieved a superior RMSE of 15.26% compared to LSR's 32.12%.

Machine learning techniques have also been used to analyze Customer Relationship Management. For instance Siddique and Wahid (2025) applied the Random Forest

algorithm to transactional data to analyze and predict customer spending. Their methodology focused on leveraging a robust, tree-based ensemble method to capture complex patterns in customer behavior. Their findings revealed that the Random Forest model offered high predictive power, effectively handling non-linear relationships within the data to deliver accurate spending forecasts. In a comprehensive study, Dang et al. (2023) investigated the feasibility of various methods for predicting customer spending based on past purchase behavior, recognizing this as crucial for business planning and budgeting. Their research specifically evaluated Linear Regression alongside more complex algorithms like Random Forest and LightGBM, establishing Linear Regression as a viable baseline model for spending prediction tasks. While their work explored advanced probabilistic models like BG/NBD and Gamma-Gamma for customer lifetime value, the inclusion and evaluation of Linear Regression validates its relevance for spending prediction problems.

Another significant contribution was made by Tsai and Lu (2009), who explored hybrid ensemble models for customer churn prediction by combining back-propagation artificial neural networks (ANN) and self-organizing maps (SOM). Their experimental results demonstrated that these hybrid models (ANN+ANN and SOM+ANN) significantly outperformed a single neural network baseline in terms of accuracy and error reduction. Further demonstrating the power of machine learning in customer behavioural prediction, Martinez et al. (2020) developed a dynamic framework to predict future customer purchases in a non-contractual setting. They engineered a set of features from historical transaction data and evaluated several state-of-the-art algorithms. Their findings revealed that the Gradient Tree Boosting method was the best-performing model, achieving an accuracy of 89% and an AUC of 0.95. Banswal et al. (2024) explicitly sought to analyze customer spending patterns and understand the significance between the demographic factors and their influence in the buying decisions made by the customers by applying several supervised machine learning methods, including Decision Trees, Random Forest, Naive Bayes, and Linear Regression, to e-commerce data.

Peker et al. (2017) conducted a study to propose a hybrid approach which predicts customers' individual purchase behaviors and reduces the limitations of these two methods by combining their advantages. The effectiveness of the proposed approach was

experimentally evaluated in the domain of supermarket shopping by using real-world data and using five popular machine learning classification algorithms including logistic regression, decision trees, support vector machines, neural networks and random forests. A key finding from their experimental results was that Logistic Regression performed better than all the other, more complex classifiers in predicting customer purchase behavior.

The strategic importance of leveraging statistical models for customer insight is demonstrated in research focused on model optimization. Rofik & Hidayat (2023) aimed to improve the accuracy of predicting customers based on purchasing behavior by optimizing a Logistic Regression model with feature selection. Their study benchmarked this optimized model against six other algorithms; K-Nearest Neighbors (KNN), Gradient Boosting, Decision Tree, Naïve Bayes, Support Vector Machine (SVM), and Random Forest. The key finding was that the optimized Logistic Regression model achieved the highest accuracy (93.18%), outperforming all the more complex alternatives.

2.3 Summary of research gaps in the related models

Although widely employed in customer analytics, Multiple Linear Regression makes strong statistical assumptions including linearity, independence of errors, homoscedasticity, and normal distribution of residuals. The studies conducted by Prathiba et al. (2024) and Dang et al. (2023) utilize Linear Regression primarily as a baseline comparator but do not thoroughly validate these fundamental assumptions, nor do they provide comprehensive diagnostic solutions for potential violations through techniques such as variable transformation or robust regression methods. Logistic regression models, as optimized by Rofik & Hidayat (2023), provide excellent classification accuracy but are fundamentally limited to categorical outcomes and cannot directly predict continuous spending magnitudes. Ensemble methods like Random Forest and Gradient Boosting, employed by Martínez et al. (2020) and Siddique & Wahid (2025), deliver high predictive accuracy but sacrificed the interpretability that is essential for actionable business insights in customer relationship management. When more flexible algorithms are introduced for comparison as seen in Prathiba et al. (2024) and Rajesh & Saravanan

(2022) the multiple linear regression model is often used as a simple benchmark rather than being systematically optimized and diagnostically validated to ensure it is the most robust linear representation of the data. This lack of a rigorously validated linear baseline makes it difficult to truly ascertain whether the increased complexity of alternative models is necessary or justified for the specific task of spending prediction.

Finally there is insufficient emphasis on the transparent communication of model characteristics through diagnostic visualization. The inclusion of residual plots, partial regression plots, and other diagnostic graphics would more prominently highlight the strengths and limitations of regression models.

Chapter 3: METHODOLOGY

3.1 Research Design

This study employed a quantitative research design, comprising predictive modeling technique to analyze customer spending. The quantitative approach was appropriate for this research as it allows for statistical analysis of numerical data and provides an objective assessment of the factors influencing spending patterns. The research focused on developing and validating a Multiple Linear Regression model to quantify the relationship between customer characteristics and spending scores.

3.2 Dataset Description

The dataset was obtained from the Kaggle website consisting of 200 rows and 10 columns. It was split into two; training and testing sets, with 80 percent allocated for training and 20 percent for testing. The training dataset was used to develop the regression model, while the testing dataset was reserved for evaluating model performance. The dataset contains nine variables as described in Table 3.1

Variable	Description
Spending Score (1-100)	A score representing the customer's spending with 1 being the least score and 100 the highest score (target variable).
Gender	Sex of the individual (Male, Female).
Age	The age of the individual.
Annual Income	Annual income of the individual.
Age Group	The categorical age group of the individual.
Estimated Savings	The estimated savings of the individual.
Credit Score	The individual's credit score.
Loyalty Years	Number of years the individual has been a customer.
Preferred Category	Preferred spending category.

Table 3. 1: Description of Model Variables

3.3 Data Pre-processing

The pre-processing of the dataset is a critical step to ensure data quality and readiness for analysis. Any missing values in the dataset were addressed using appropriate imputation techniques, such as mean imputation for numerical variables and mode imputation for categorical variables. Categorical variables including Gender, Age Group, and Preferred Category were encoded into numerical forms using dummy variables to ensure compatibility with the regression model. Continuous variables such as Age, Annual Income, Estimated Savings, Credit Score and Loyalty Years were standardized to bring them to a uniform scale, facilitating better model performance and interpretability. Outliers in numerical variables were identified using methods such as the interquartile range. Decisions regarding outlier treatment was made based on their impact on model performance with options including capping extreme values or applying transformations to minimize distortion of the results.

3.4 Exploratory Data Analysis (EDA)

Following data pre-processing, exploratory data analysis was conducted to understand the characteristics of the dataset, detect potential issues in the dataset, identify underlying patterns and correlations among variables. This involved computing descriptive statistics and data visualization.

Descriptive statistics included measures of central tendency, measures of dispersion and correlation analysis. For data visualization several plots are used. A Density plot was used to illustrate the probability distribution of the dependent variable. A box Plot to show the median, inter quartile range and outliers among different subgroups under preferred category. Scatter plots were widely used to display relationships between variables and a bar Plot was utilized to compare the spending scores between the different loyalty years.

3.5 Model Assumptions

3.5.1 Linearity check

Linearity means the relationship between predictors and the dependent variable is a straight line. Linearity check involved visualizing a Residuals plot.

3.5.2 Independence check

Independence refers to a lack of relationship (autocorrelation) between error terms. Durbin-Watson test was utilised to check for autocorrelation.

$$d = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$$

Where: e_t are residuals

3.5.3 Homoscedasticity check

Homoscedasticity means the variance of the residuals is constant across all levels of the independent variables. A residual plot of absolute residuals was fitted to observe the spread of the residuals.

An additional test method used is the Breusch-Pagan test.

$$LM = n \times R_{residuals}^2$$

Where

$R_{residuals}^2$: comes from regressing squared residuals on predictors.

LM Statistic follows chi-square distribution

Significant p-value indicates heteroscedasticity. ($P < 0.05$)

3.5.4 Normality check

Normality refers data following a normal distribution. This was visualised using a Q-Q plot.

3.5.5 Multicollinearity check

Multicollinearity is the correlation between independent variables. The test method used to check for multicollinearity is Variance Inflation Factor (VIF). VIF measures how much multicollinearity inflates the variance of a regression coefficient making it unreliable.

$$VIF_j = \frac{1}{1 - R_j^2}$$

Where

R_j^2 : is R-squared from regressing predictor j on all other predictors.

VIF < 5 indicates no multicollinearity, VIF > 5 suggests moderate multicollinearity and VIF > 10 indicates severe multicollinearity.

3.6 Multiple Linear Regression

Multiple Linear Regression (MLR) is a statistical technique that extends simple linear regression to model the relationship between a single continuous dependent variable and two or more independent variables also known as predictors or regressors. The primary goal of MLR is to find a linear equation that best predicts the value of the dependent variable based on the values of the independent variables in the observed data.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

Where:

- Y : The dependent variable
- X_1 through X_k : The independent variables.
- β_0 : The expected value of Y when all independent variables are zero.
- β_1 through β_k : The regression coefficients, representing the partial effect of each independent variable on Y while holding the other predictors constant.
- ε : The residual error.

Observation Number	Response Y	Response Y
1	y_1	$X_{11}, X_{12}, \dots, X_{1k}$
2	y_2	$X_{21}, X_{22}, \dots, X_{2k}$
$\vdots$	$\vdots$	$\vdots$
n	y_n	$X_{n1}, X_{n2}, \dots, X_{nk}$

Table 3. 2: Multiple Linear Regression Data Structure

Matrix Form of the Model

For n observations, the model can be written concisely in matrix form:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

$$Y = X\beta + \varepsilon$$

Where:

- Y : Vector of n observed values of the dependent variable.
- X : $n \times (k + 1)$ design matrix of independent variables.
- β : Vector of $k + 1$ regression coefficients to be estimated.
- ε : Vector of n error terms.

3.6.1 Parameter Estimation

The Ordinary Least Squares (OLS) method is used to estimate the regression coefficients β by minimizing the sum of squared errors (SSE).

Sum of Squared Errors (SSE):

$$SSE = \varepsilon^T \varepsilon = (y - X\beta)^T (y - X\beta)$$

To obtain the parameters the SSE is minimized by taking its derivative with respect to β , setting it to zero, and solving for the coefficients.

Expand the SSE:

$$\begin{aligned}SSE &= (y - X\beta)^T(y - X\beta) \\&= y^T y - 2\beta^T X^T + \beta^T X^T X\beta\end{aligned}$$

Differentiate with respect to β :

$$\frac{\partial SSE}{\partial \beta} = -2X^T y + 2X^T X\beta$$

Set the derivative to zero and solve:

$$0 = -2X^T y + 2X^T X\beta$$

$$\beta = (X^T X)^{-1} X^T y$$

The OLS Estimator:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

3.7 Goodness of fit of the Model

Goodness of fit measures how well a statistical model matches a set of observed data, essentially quantifying the discrepancy between the models prediction and the actual data

The following key metrics were used to evaluate the goodness of fit of the fitted Multiple Linear Regression model:

3.7.1 R-squared (R2)

R-squared or the coefficient of determination, is a statistical measure representing the proportion of the variance in the dependent variable(y) that is predictable from the independent variables(X1, X2 . . . , Xn). R-squared typically ranges from 0 to 1 or 0% to 100% and a higher value indicates that a larger percentage of the variability is explained

by the model suggesting a better fit. A key limitation of the R-squared value is that it automatically increases or stays the same every time you add a new independent variable to a model even if that variable provides no meaningful information, which can make a complex over fit model look deceptively good. To calculate R squared you first need the total sum of squares (SST) which measures the total variation in the observed dependent variable values from their mean and the residual sum of squares (SSR) which measures the sum of the squared differences between the observed and predicted values.

$$R^2 = 1 - \frac{\text{Sum of Squared Residuals (SSR)}}{\text{Total Sum of Squares (SST)}}$$

Where:

- $R^2 = 1$: Perfect fit
- $R^2 = 0$: Model explains no variance

3.7.2 Adjusted R-squared

Adjusted R-squared is a modified version of R-squared that accounts for the number of predictors in the model. This metric addresses the bias of the standard R-squared by penalizing the addition of unnecessary independent variables. The adjusted R-squared increases only if the new predictor improves the model's performance more than would be expected by chance otherwise it will decrease. It is interpreted as the proportion of total variance that is explained by the model.

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1}$$

Where:

- n: Number of observations
- k: Number of predictors

3.8 Evaluation of Models' predictive Accuracy

Evaluating the models' predictive accuracy involves using specific metrics and techniques to quantify how well its prediction align with actual outcomes using unseen data and identify potential issues like overfitting or under fitting. This crucial step in the data analysis ensures the model is reliable and generalizable not only to the training data but also the testing dataset. The following metrics were used to help in measuring how well the fitted multiple linear regression model is able to predict continuous outcomes.

3.8.1 Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) measures the average magnitude of the errors between actual observed values and predicted values without regard for the direction (positive or negative) of the error. It is also known for being more robust to outliers compared to RMSE because it weighs all errors linearly meaning a very large error doesn't disproportionately skew the average error as much.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where:

- n = the total number of observations
- Y_i = the actual observed value
- $\hat{y}_i$ = the predicted value from the model
- $y_i - \hat{y}_i$ = the absolute error between the actual and predicted values

3.8.2 Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) is a crucial metric for evaluating the absolute fit of a regression model to the data. It is fundamentally defined as the square root of the average of the squared errors effectively representing the standard deviation of the residuals. This measurement indicates how tightly the observed data points cluster around the models predicted values. A highly useful property of RMSE is that it is expressed in the same units as the dependent variable, which makes the magnitude of the typical

prediction error easy to interpret. Lower values of RMSE indicate better fit. RMSE is a good measure of how accurately the model predicts the response. It's the most important criterion for fit if the main purpose of the model is prediction.

$$RMSE = \sqrt{MSE}$$

CHAPTER 4: ANALYSIS AND RESULTS

4.1 Introduction

This chapter presents the results and findings from the analysis of customer spending prediction using multiple linear regression model. The analysis includes exploratory data analysis, model implementation and performance evaluation. Python Programming Language was used to analyze the dataset.

4.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the characteristics of the dataset, identify patterns and detect potential issues such as missing values, outliers and correlations among variables. The next step was to select the relevant features that are most predictive of customer spending. This involved correlation analysis, descriptive statistics and domain expertise.

Data analysis and visualization play a critical role in understanding the underlying patterns and relationships in the data and identifying potential outliers and anomalies. Exploratory Data Analysis on the customer spending dataset was to gain insights into the distribution of features, identify correlations between different variables, and visualize the relationships between features and the target variable. Descriptive statistics provide an overview of the dataset's characteristics.

The key statistical measures are shown in table 4.1.

	Age	Annual Income	Spending Score	Estimated Savings	Credit Score	Loyalty Years
count	200.000000	200.000000	200.000000	200.000000	200.000000	200.000000
mean	38.850000	60.560000	50.200000	40.247950	743.680000	5.930000
Std	13.969007	26.264721	25.823522	21.333448	155.176727	1.570808
min	18.000000	15.000000	1.000000	6.460000	300.000000	2.000000
Q1	28.750000	41.500000	34.750000	28.800000	697.000000	5.000000
Median	36.000000	61.500000	50.000000	36.410000	833.000000	6.000000
Q3	49.000000	78.000000	73.000000	44.967500	850.000000	7.000000
Max	70.000000	137.000000	99.000000	120.560000	850.000000	9.000000

Table 4. 1: Descriptive Statistics of numerical features in the Dataset

The correlation analysis heat map (figure 4.1) revealed significant and sometimes unexpected relationships between customer characteristics and spending score. Annual income demonstrated a very strong positive correlation with estimated savings ($r = 0.81$) and credit score ($r = 0.75$) indicating that wealthier customers tend to save more and maintain better credit. However, in a surprising departure from conventional assumptions, annual income showed virtually no correlation with spending score ($r = 0.0099$), suggesting that financial capacity does not directly translate to higher spending behavior in this customer sample. Instead, spending score showed moderate positive correlations with loyalty years ($r = 0.53$) and moderate negative correlations with both age ($r = -0.33$) and estimated savings ($r = -0.52$), revealing that younger, loyal customers who save less tend to be the highest spenders. This pattern suggests a clear behavioral trade-off where customers who spend heavily tend to save less and vice versa. Additionally, the strong positive correlation between age and loyalty years ($r = 0.61$) indicates that customer retention increases with age, while the negative correlation between spending score and credit score ($r = -0.30$) suggests that more financially responsible customers exercise greater spending restraint.

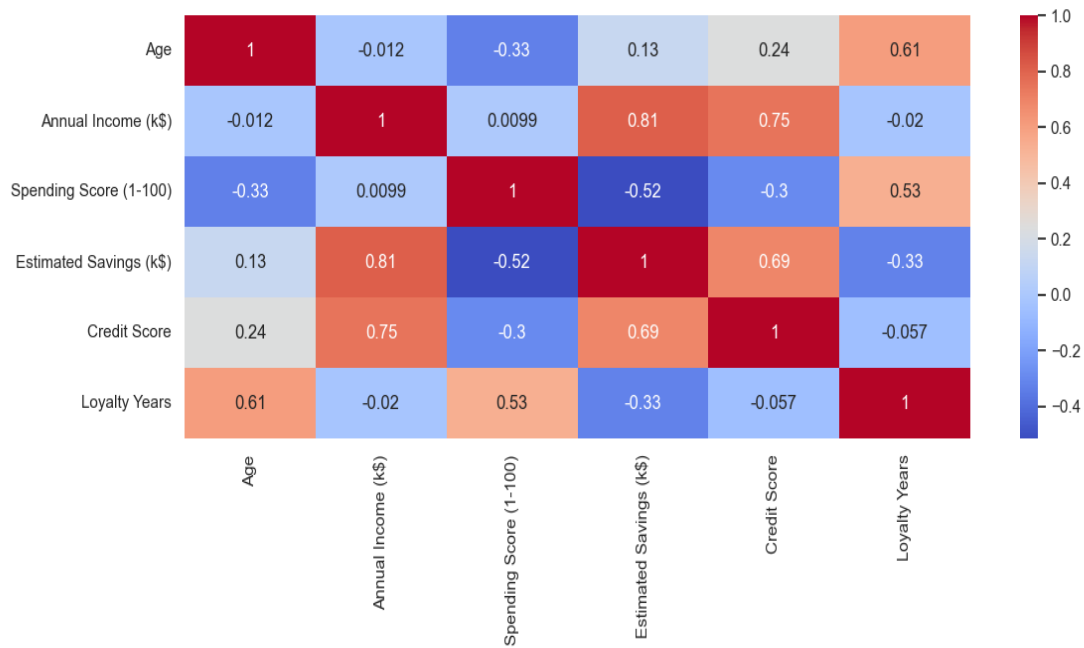


Figure 4. 1: Correlation Heat map for Numerical Variables

This density plot (figure 4.2) illustrates the probability distribution of the Spending Score (1-100) among customers. The histogram overlaid with a kernel density estimation (KDE) curve, provides a smooth estimate of the distribution. The plot exhibits a unimodal shape, with the highest density concentrated around a spending score value of 50. This suggests that most customers' spending score fall within this range. To further emphasize on this, the KDE curve shows a prominent roughly symmetrical peak centered on a score of 50, indicating that the most common spending behavior by customers is moderate, falling squarely in the middle of the spender spectrum. It is also notable that a proportion of customers spend significantly above the mean. The smooth, bell-shaped taper on both sides towards the extremes scores near 0 and 100, reveals that exceptionally low and high spenders are progressively less frequent. The smooth, bell-shaped pattern further indicates the absence of distinct subgroups based solely on spending score. This unimodal distribution is advantageous for regression-based predictive models, as it implies a continuous and centralized relationship with predictor variables.

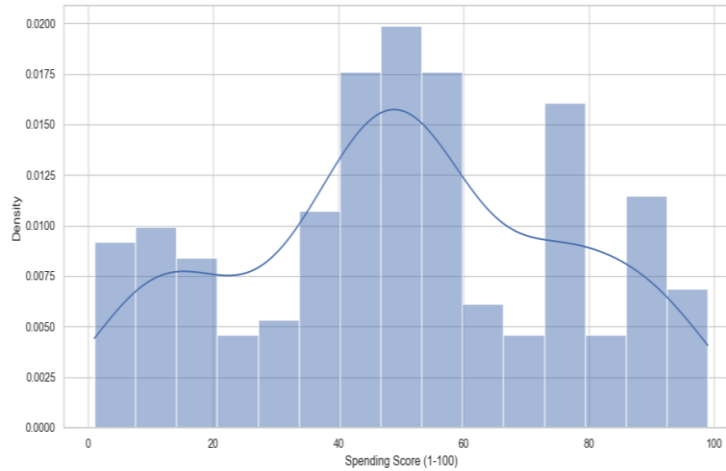


Figure 4. 2: Distribution of spending score

Figure 4.3 analyses the relationship between customers' Preferred Product Category and their Spending Score. The median spending score for Budget is the least at 28, Electronics and Fashion seem to be at the middle with 44.5 and 49 respectively while Luxury has the highest median at 82.5. This suggest that higher spenders typically comprise of people who prefer luxury while those who prefer budget spend the least. Additionally, the interquartile range (IQR) for luxury is the most compact, spanning from around 71 to 91, followed by budget which is between 3 and 36. This highlights a clear pattern both where both have minimal variability in their spending but from two opposite ends of the spending score. The IQR for electronics and fashion are the most spread out spanning from 1 to 55 indicating that they experience the most variability spending. The whiskers in the boxplot shows the presence of outliers in all the four groups. The maximum spending score for luxury is 99, fashion 69, budget 66 and electronics 60. Despite budget being a conservative category there are individuals who still spend more than expected. The minimum spending score for luxury is 71, budget 3, fashion 1 and electronics 1. These findings emphasize that the categorical variable preferred category is a key determining factor of spending.

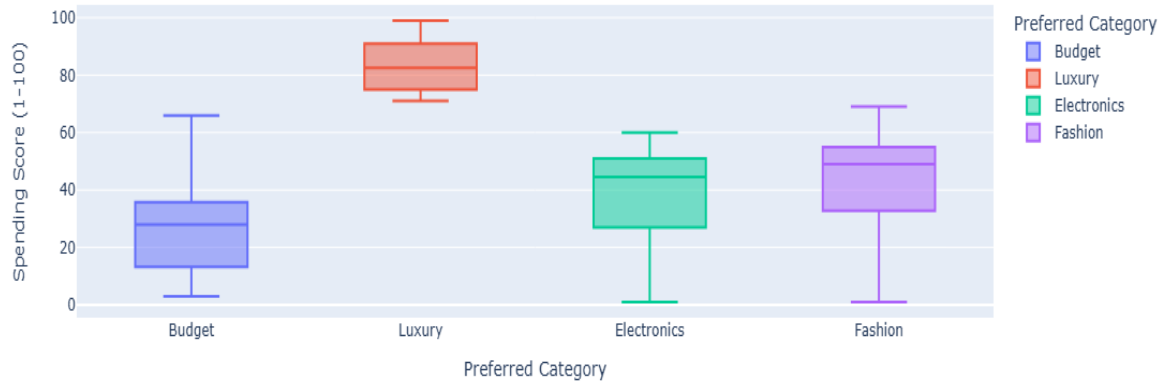


Figure 4. 3: Box plot of preferred category and Spending Score

Figure 4.4 illustrates the relationship between spending score and age group, revealing a clear non-linear pattern. Spending scores are relatively high among customers aged 18 to 25 and reach their peak in the 26 to 35 age group, which likely represents individuals in early career stages with increasing disposable income and fewer long-term financial commitments. A noticeable decline in spending is observed in the 36 to 50 and 51 to 65 age groups, possibly reflecting greater financial responsibilities such as housing, education and long-term savings. A slight increase in spending is observed again in the 65 and above group, which may be associated with retirement spending or a focused customer base with greater financial freedom. This curvilinear pattern suggests that using age brackets helps to capture the non-linear influence of customer age on spending propensity

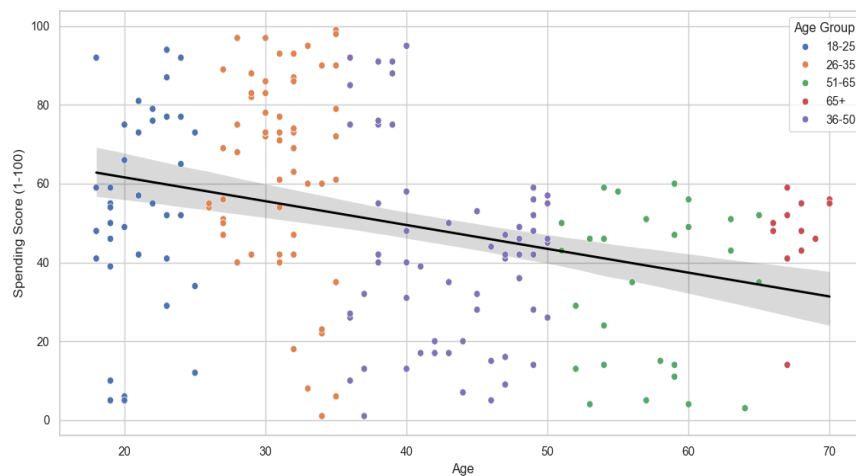


Figure 4. 4: Scatterplot of Spending score and Age

Figure 4.5 shows a very weak relationship between annual income and spending score, as indicated by the nearly horizontal regression line. This suggests that income is not a strong determinant of spending behavior in this dataset. Luxury-oriented customers consistently exhibit high spending scores across a wide range of income levels, while budget-oriented customers remain concentrated at low spending scores regardless of income. Customers who prefer fashion and electronics are mainly associated with moderate spending levels, with some variation across income. Overall, the figure confirms that customer spending behavior is influenced more by preferences and behavioral factors than by income alone.

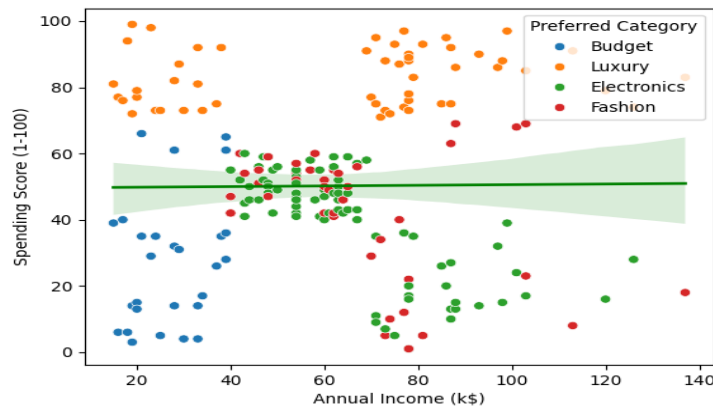


Figure 4. 5: Scatterplot of Spending score and Annual Income

Figure 4.6 examines the relationship between estimated savings and spending score, with observations categorized by preferred product category. The negative slope of the fitted trend line indicates a clear inverse relationship, whereby customers with higher estimated savings tend to exhibit lower spending scores. This pattern suggests a behavioral trade-off between saving and spending, with more financially conservative customers prioritizing savings over consumption. Customers who prefer luxury products (orange) tend to cluster at higher spending scores and lower to moderate savings levels, while budget-oriented customers (blue) are concentrated at low levels of both spending and savings. Fashion (red) and electronics (green) customers generally occupy intermediate ranges of both variables.

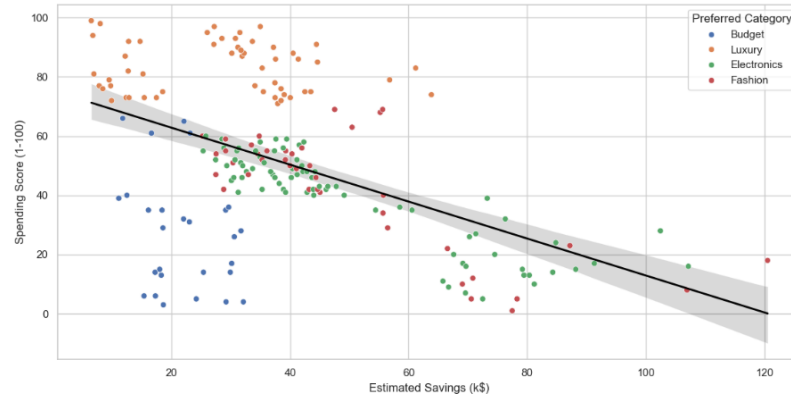


Figure 4. 6: Scatterplot of Spending score and estimated savings

Figure 4.7 illustrates the relationship between Credit Score and Spending Score, with points categorized by preferred category. The trend line indicates a slight negative trend, suggesting that as credit scores increase, spending scores marginally decrease. Customers with a preference for Luxury items form a distinct cluster in the high spending score range approximately 50-100 across all credit scores, while Budget oriented customers are densely concentrated in the low spending score range 0-50, regardless of their credit health. Electronic and fashion lovers have a healthy credit score of above 550 and moderate their spending to a maximum of 60.

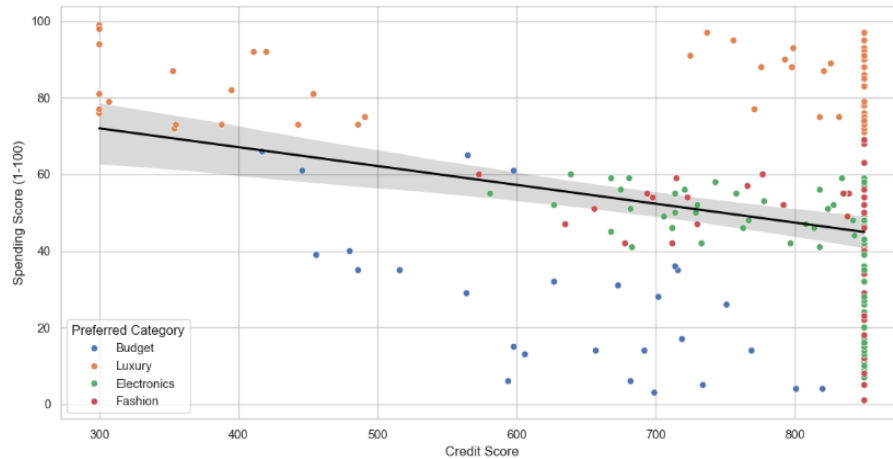


Figure 4. 7: Scatterplot of Spending score and credit score

The bar plot Figure 4.8 tracks how the Average Spending Score changes with a customer's Loyalty years. The trend reveals a clear positive relationship that appears to plateau. There is a steep increase in average spending score from new customers (2 years)

to those with moderate tenure (5-6 years), indicating that loyalty is earned and rewarded with greater spending. However, after approximately 7 years, the average score stabilizes, forming a plateau. This suggests a point of diminishing returns on loyalty. The initial years are crucial for building spending engagement, but after a certain threshold, spending habits become entrenched and no longer increase with time. For business strategy, this highlights the importance of programs targeted at customers in the 2-6 year loyalty window to accelerate them into high-spending tiers.

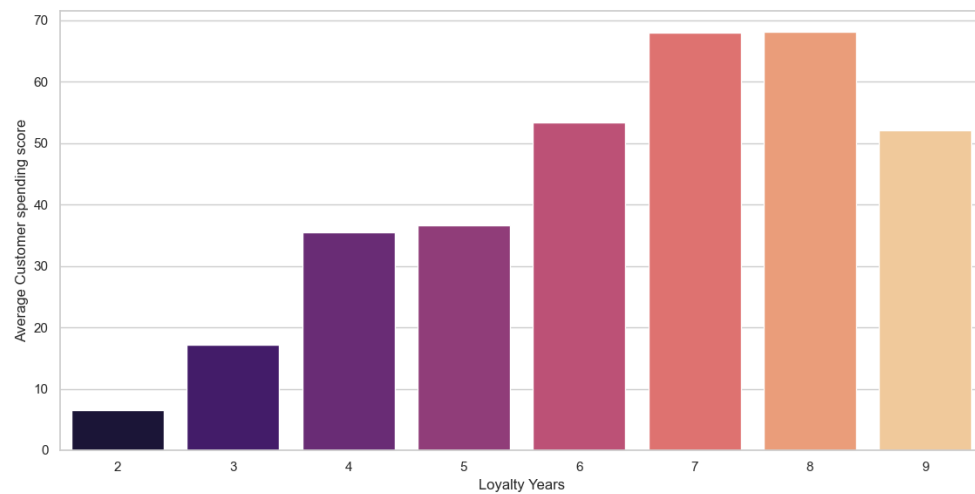


Figure 4. 8: bar plot of Average spending score and loyalty years

4.3 Multiple Linear Regression

A linear regression model was fitted to predict medical insurance charges based on annual income, age group, credit score, loyalty years and preferred category. The resulting OLS summary is shown below:

	Coefficient	Std Error	t-value	p-value	[0.025	0.975]
Const	63.5457	4.855	13.090	<0.001	53.952	73.061
Annual Income	-1.2575	1.332	-0.944	0.347	-3.890	1.354
Credit Score	1.3543	1.213	1.117	0.266	-1.043	3.731
Loyalty Years	22.1655	1.284	17.266	<0.001	19.629	24.682
Age Group_26-35	-11.6917	2.524	-4.632	<0.001	-16.680	-6.744
Age Group_36-50	-28.4230	3.787	-7.505	<0.001	-35.907	-21.001
Age Group_51-65	-48.9520	4.792	-10.216	<0.001	-58.421	-39.560
Age Group_65+	-61.0316	5.826	-10.475	<0.001	-72.545	-49.612
Preferred Category Electronics	5.2063	3.001	1.735	0.085	-0.723	11.087
Preferred Category Fashion	7.4671	3.578	2.087	0.039	0.397	14.479
Preferred Category Luxury	21.0046	4.151	5.060	<0.001	12.801	29.141
Gender male	0.4460	1.348	0.331	0.741	-2.218	3.110
R-squared:	0.918					
Adjusted R-squared:	0.912					
F-statistic:	154.5					
Prob (F-statistic):	< 0.001					

Table 4. 2: The OLS results for the multiple linear regression model

The model explains approximately 91.8% of the variation in customer spending score, with an R-squared value of 0.918. The adjusted R-squared of 0.912 indicates strong explanatory power of the selected predictors. All categorical variables were represented using dummy variables so that the effect of each category could be compared relative to a reference group. The reference categories were customers aged 18–25, those whose preferred spending category is Budget and female customers. Loyalty years emerged as

the strongest predictor of spending score with a unit increase increasing the score by approximately 22.1655, indicating that customers who have maintained longer relationships with the business tend to exhibit substantially higher spending scores compared to newer customers. Age group also has a significant effect with all older age groups showing lower spending scores compared to customers aged 18–25. Customers aged 26–35 spend approximately 11.69 points less than those aged 18–25, while customers aged 36–50 spend about 28.42 points less. Similarly customers aged 51–65 spend approximately 48.95 points less and those aged 65 and above spend about 61.03 points less than customers in the 18–25 age group. With respect to product preferences, customers who prefer Fashion spend approximately 7.47 points more than customers who prefer Budget items, while those who prefer Luxury products spend about 21.00 points more than the Budget category. Although not statistically significant the Electronics category shows an increase of approximately 5.21 points compared to the Budget category. In contrast gender, annual income and credit score are not statistically significant predictors at the 5% level. This suggests that, after controlling for behavioural and demographic factors, financial capacity and gender differences do not meaningfully influence customer spending in this context.

After removing the statistically insignificant predictors, the reduced model continues to demonstrate very strong explanatory power. The model explains approximately 91.7% of the variation in customer spending scores ($R^2 = 0.917$). More importantly, the adjusted R-squared increased slightly to 0.913, indicating a small improvement in model quality after eliminating unnecessary variables.

	Coefficient	Std Error	t-value	p-value	[0.025	0.975]
Const	63.8995	3.018	21.176	<0.001	57.937	69.862
Loyalty Years	22.1445	1.092	20.275	<0.001	19.987	24.303
Preferred Category Electronics	4.6004	2.110	2.181	0.031	0.432	8.769
Preferred Category Fashion	7.4076	2.575	2.876	0.005	2.319	12.496
Preferred Category Luxury	19.8343	2.801	7.081	<0.001	14.300	25.368
Age Group_26-35	-11.6861	2.059	-5.676	<0.001	-15.754	-7.618
Age Group_36-50	-27.7892	2.628	-10.576	<0.001	-32.981	-22.597
Age Group_51-65	-48.0947	3.627	-13.259	<0.001	-55.261	-40.928
Age Group_65+	-59.7769	4.799	-12.456	<0.001	-69.259	-50.295
R-squared:	0.917					
Adj. R-squared:	0.913					
F-statistic:	209.4					
Prob (F-statistic):	< 0.001					

Table 4. 3: The OLS results for the statistically significant variables

4.4 Model Assumptions

Linearity Check

In figure 4.9, the residuals appear to be randomly scattered around the zero line across the range of fitted values, suggesting that the linear model captures the general relationship between the predictors and the response variable. Although the LOWESS curve shows slight curvature, the deviations are relatively small and do not indicate a strong systematic pattern. This suggests that the assumption of linearity is reasonably satisfied for the fitted regression model.

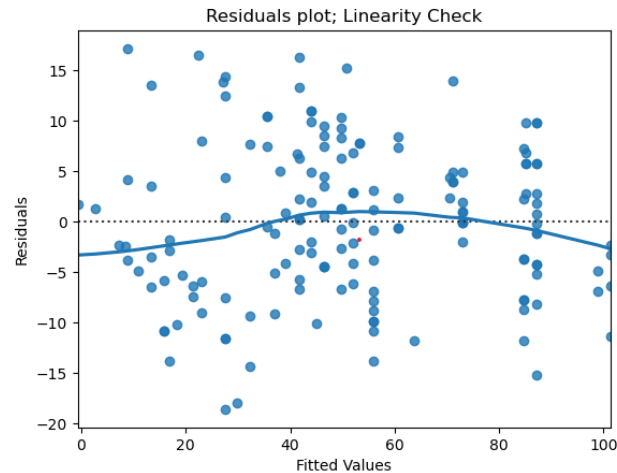


Figure 4. 9: Linearity Check

Normality Check

Figure 4.10, a Kernel Density Estimate of the residuals was used to assess whether the residuals of the regression model are normally distributed. The residuals in this graph appear very close to a normal distribution, the curve is symmetrical and centered around zero indicating that the model's errors are balanced on both positive and negative sides. The distribution is unimodal displaying a single distinct peak near zero suggesting the residuals likely originate from a single underlying process without major, distinct subgroups. The tails appear consistent with a normal curve, with residuals generally falling within the -20 to +20 range.

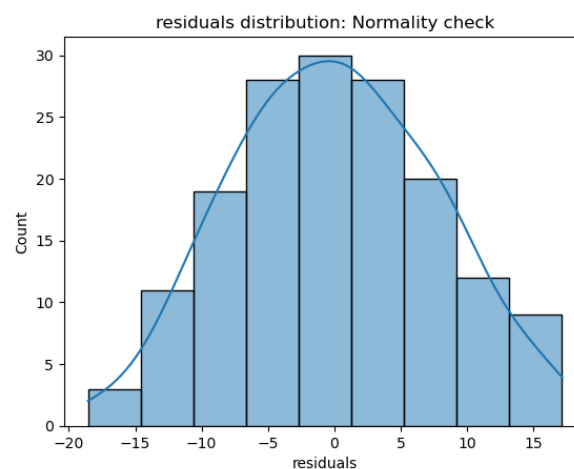


Figure 4. 10: Normality Check

The Q–Q plot figure 4.11 show that the data points align very closely with the red straight diagonal line, which is the primary indicator of a normal distribution. There is no significant S-shapes, curves or major deviations at the tails, which means the residuals are symmetrical and lack extreme outliers or significant skewness

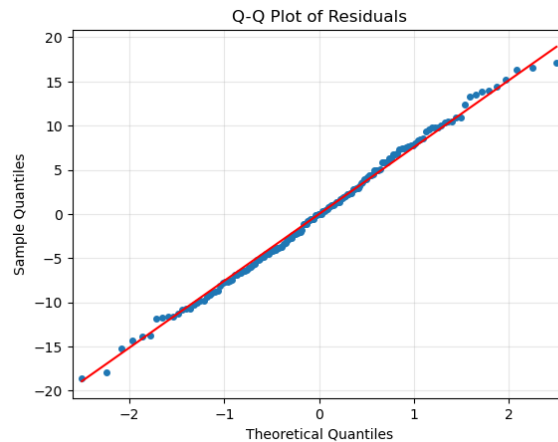


Figure 4. 11: Normality of the residuals

The Shapiro–Wilk test produced a statistic of 0.9932 with a p-value of 0.6627, indicating no statistically significant departure from normality.

Homoscedasticity Check

In figure 4.12, the scatterplot of residuals versus fitted values shows no clear funnel shape, but there is noticeable spread variation across the range. Residuals appear more scattered at lower fitted values and slightly compressed at higher fitted values. This suggests potential mild heteroscedasticity,

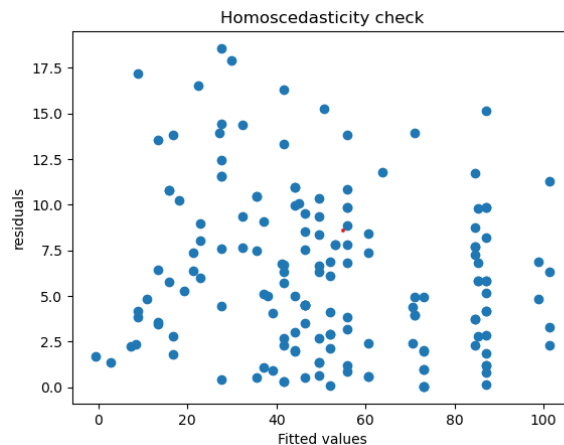


Figure 4. 12: Homoscedasticity check

Furthermore, the Breusch-Pagan test yielded a p value of 0.0297, confirming the presence of heteroscedasticity at the 5% significant level. Given the mild nature of the detected heteroscedasticity, heteroscedasticity-consistent (HC3) robust standard errors were employed to ensure valid statistical inference.

Independence of Errors

The Durbin-Watson statistic of 2.1880 is close to ideal value of 2, suggesting little to no autocorrelation in the residuals. These results indicate that the independence assumption is satisfied.

Multicollinearity

All the VIF values except for constant term are less than 5 it indicates that no multicollinearity, that is there is no significant relationships among the predictors as shown below.

Feature	VIF
Const	23.969462
Loyalty Years	3.244847
Preferred Category Electronics	2.746015
Preferred Category Fashion	2.590832
Preferred Category Luxury	4.336725
Age Group 26-35	2.397172
Age Group 36-50	3.946795
Age Group 51-65	4.107233
Age Group 65+	3.218141

Table 4. 4: Variance Inflation Factor (VIF) for Each Feature

4.5 Goodness of fit of the Model

The table below shows the R-squared and R-squared values for both models.

Metric	Model 1	Model 2
R-squared	0.918	0.917
Adjusted R squared	0.912	0.913

Table 4. 5: Goodness of fit metrics

The comparison between Model 1 and Model 2 indicates a slight difference in overall model fit. Model 1 explains approximately 91.8% of the variation in Spending Score while model 2 explains approximately 91.7% of the variation. While this suggests strong explanatory power, R-squared alone does not account for the number of predictors included in the model therefore greater emphasis is placed on the adjusted R-squared which penalizes the inclusion of unnecessary variables. The adjusted R-squared for Model 2 (0.913) is slightly higher than that of Model 1 (0.912), indicating that the reduced model achieves comparable explanatory strength while using fewer predictors. This improvement indicates that the variables removed from Model 1 were not contributing meaningful explanatory information to the model. Consequently, the adjusted R-squared provides stronger evidence that Model 2 represents a more efficient specification. Overall, the goodness-of-fit measures confirm that the reduced model maintains a high explanatory power of 91.7% while maintaining model simplicity.

4.6 Evaluation of Models' Performance

The dataset was divided into training and test sets using a standard 80–20 split, where 80% of the observations were used to train the regression model and the remaining 20% were reserved for evaluating predictive performance on unseen data. This proportion provides an appropriate balance between model learning and validation, ensuring sufficient data for pattern estimation while maintaining a reliable test set for performance assessment. A fixed random state was applied to ensure reproducibility. Both Model 1 (full model) and Model 2 (reduced model excluding statistically insignificant variables) were trained on the training dataset and evaluated on the test dataset using Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

	MAE	MSE	RMSE
Model 1	7.1736	78.8496	8.8797
Model 2	7.1085	78.1854	8.8422

Table 4. 6: Evaluation metrics

The marginally lower error metrics for Model 2 indicate a very small improvement in predictive performance. However, the differences between the two models are minimal suggesting that the excluded variables do not meaningfully enhance prediction accuracy. Because RMSE penalizes larger errors more heavily than MAE, the nearly identical RMSE values indicate that both models perform similarly in handling larger deviations between predicted and actual Spending Scores. Likewise, the almost identical MAE values suggest that the average absolute prediction error is effectively the same across both models. On average, the models' predictions deviate from actual Spending Scores by approximately 7.1 to 8.8 points, which is relatively small given the 1 to 100 scale of the dependent variable. Overall, the predictive accuracy analysis demonstrates that removing statistically insignificant variables does not compromise the model's ability to generalize to unseen data. The reduced model achieves virtually the same prediction performance while maintaining greater simplicity.

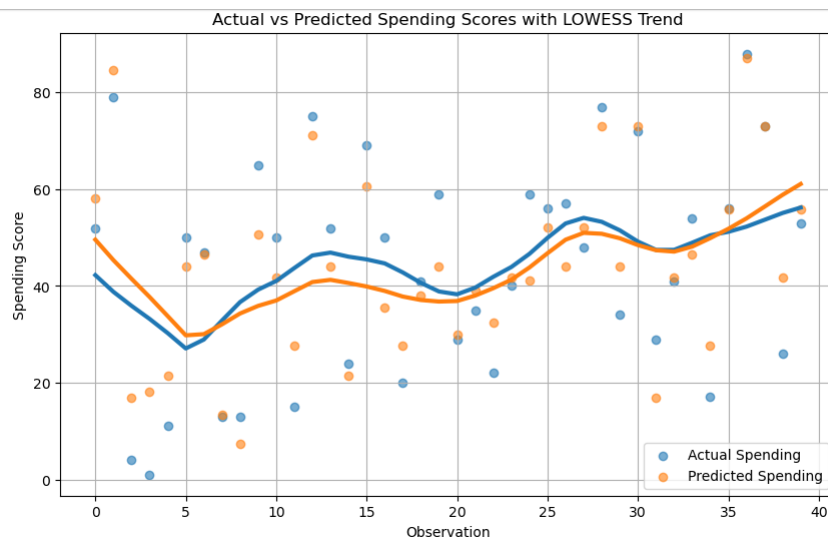


Figure 4. 13: Scatterplot of Predicted values

CHAPTER 5: CONCLUSION AND RECOMMENDATION

5.1 Conclusion

This study modelled and analyzed customer spending scores using Multiple Linear Regression based on demographic, financial and behavioural characteristics obtained from a customer analytics dataset. The analysis demonstrated that the regression model successfully established statistical relationships between selected customer attributes and spending behaviour. Diagnostic assessments indicated that the key regression assumptions, including linearity, normality of residuals, independence of errors, homoscedasticity and the absence of severe multicollinearity, were reasonably satisfied. These results confirm that the fitted regression model is statistically appropriate for both inference and prediction within the context of the dataset. The final reduced model exhibited strong explanatory capability, explaining approximately 91.7% of the variation in customer spending scores ($R^2 = 0.917$; Adjusted $R^2 = 0.913$). The small difference between R-squared and adjusted R-squared indicates that the selected predictors contribute meaningful information to the model without introducing unnecessary complexity. In addition, the model demonstrated good predictive performance when evaluated on the test dataset, producing relatively low prediction errors (MSE = 72.79, RMSE = 8.53 and MAE = 6.84) relative to the 1–100 spending score scale. The results further revealed that loyalty years, age group and preferred spending category significantly influence customer spending behaviour. In contrast, gender, annual income and credit score were not statistically significant predictors within this dataset. The removal of these non-significant variables did not reduce the explanatory power or predictive accuracy of the model, indicating that behavioural and demographic characteristics play a more substantial role in explaining customer spending patterns than purely financial attributes in this context. Overall, the findings demonstrate that Multiple Linear Regression provides a reliable and interpretable statistical framework for explaining and predicting customer spending behaviour and the model developed in this study offers useful insights that can support data-driven decision making in customer analytics.

5.2 Recommendations

Future implementations of customer spending prediction models should be integrated into customer relationship management systems to enable real-time and periodic analysis of customer behavior. Embedding predictive models within operational systems allows businesses to identify high-value customers, monitor spending trends and allocate resources more efficiently using data-driven insights. Although Multiple Linear Regression demonstrated strong explanatory power in this study, future research should incorporate regularization techniques such as Ridge and Lasso regression to improve coefficient stability and address potential multicollinearity. The use of k-fold cross-validation is also recommended to strengthen model generalizability beyond a single train-test split. Given the mild heteroscedasticity observed, approaches such as Weighted Least Squares (WLS) may be explored to better handle non-constant error variance. Additionally, investigating interaction effects, non-linear transformations and applying feature engineering or dimensionality reduction techniques could further enhance model performance and efficiency. To improve practical applicability, organizations should complement predictive modeling with interactive dashboards for monitoring performance. Expanding datasets to include more diverse customer profiles and regularly retraining the model will help maintain accuracy and ensure continued relevance as customer behavior evolves over time.

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